Dev Chauhan

Ahmedabad, Gujarat, India | +91 9662908809 | chauhandev591@gmail.com | LinkedIn | Github

Profile

(click here to view portfolio)

Computer Science Engineering student at Nirma University (**Third year, B.Tech, Class of 2028**) with hands-on experience designing, building, and testing **full-stack, AI-integrated systems** — from hybrid edge-cloud architectures to ML-driven monitoring pipelines. Strong foundation in **data structures, algorithms, and object-oriented design**, with hands-on experience in **NLP, generative AI, prompt engineering, and agentic AI tooling (Claude Code, MCP)**. Proven collaborator across hackathons and academic research, with a paper selected for publication at RIACT 2026.

Skills

Programming Languages: C, C++, Python, JavaScript, TypeScript, Java, Node, React, Tailwind, MATLAB, MongoDB, SQL/NO SQL.

Software Engineering & ML: Machine Learning (Isolation Forest, LSTM, CNN-LSTM), NLP, SQL & Data Pipelines (SQLite), Express.js, RESTful API Design, Socket.io, Mongoose/Sequelize (dual-DB), Three.js, Chrome Extension Development (Manifest V3), Git/GitHub.

Agentic AI & MLOps: Claude Code, Model Context Protocol (MCP), Agent Skills, Prompt Engineering, Hugging Face Transformer, Agentic coding

Experience

Team Arrow (Robotics Club) — **Nirma University**
Technical Team Member — *Aug 2024 – Nov 2024*

- Engineered autonomous navigation software for a **distributed**, drone-based robotics platform, integrating the Ardupilot flight stack with custom communication modules for real-time autonomous control.
- Built, tested, and debugged embedded computer-vision and sensor pipelines on Jetson Nano and Raspberry Pi (Pi HQ Cameras), strengthening systems-level engineering across hardware-software interfaces.

Projects & Publication

ReLoop | Carbon-Aware B2B Circular Packaging Exchange

- Architected a TypeScript/Node.js **modular monolith backend** (Express, Mongoose + Sequelize dual-DB) spanning 24 domain modules across 8 decoupled tiers, paired with a React 19 + **Three.js** frontend for real-time 3D load visualization.
- Engineered a deterministic **Scope 3 carbon accounting engine** computing net CO2e savings and a geographic **carbon break-even radius** per listing, automatically suppressing carbon-negative trade recommendations.
- Designed a bidirectional **multi-criteria hybrid matching engine** (5-vector weighted scoring across semantic, grade, price, carbon, and timing signals) alongside a multi-stop **vehicle-routing and 3D bin-packing solver** to optimize freight load factors.

VeriPhi | Browser-Integrated AI Trust Intelligence Layer

- Architected a **Chrome extension (Manifest V3)** that verifies LLM output in real time, decomposing streaming text into atomic claims via NLP-based sentence-boundary detection.
- Designed a **hybrid edge-cloud, distributed AI architecture**, routing PII-sensitive processing on-device (Gemini Nano) and complex reasoning to the cloud (Gemini 1.5 Flash) for low-latency, privacy-preserving inference.
- Integrated **Vertex AI Grounding** to cross-reference AI-generated claims against live search data, eliminating knowledge-cutoff errors and improving system reliability.

MERIDIAN | Multi-Asset Portfolio Optimization & Risk Engine

- Engineered a **multi-asset quantitative optimization engine** solving 3 allocation strategies (Minimum Variance, Markowitz Max Sharpe, Equal Risk Parity) via SciPy SLSQP, achieving mathematical convergence across 12-asset universes in **<50ms** execution time.
- Built a **3-tier resilient data pipeline** (yfinance → SQLite TTL cache → CSV fallback) reducing market data fetch latency from ~2.4s to **<15ms**, and implemented a **friction-adjusted backtesting engine** modeling asset drift turnover, rolling Sharpe ratios, and volatility quantile stress testing across multi-year horizons.

Conference Paper — Autonomous Disaster Management Robot (International) — **RIACT 2026**

- Designed a fully autonomous robotic system, published as "Development and Application of a Fully Autonomous Robot for Disaster Management and Humanitarian Purposes," to navigate hazardous, post-disaster environments and deliver humanitarian aid without human intervention.
- Engineered fail-safe navigation logic to eliminate physical risk exposure for human rescuers operating in unknown environments.

Education

Nirma University — **Ahmedabad, Gujarat**
B.Tech in Computer Science and Engineering - Pursuing — *Aug 2024 – July 2028*

Best High School — **Ahmedabad, Gujarat**
Higher Secondary Certificate (12th Grade), 83% — *2009 – 2024*

Co-Curricular

- **Hackathons/Competition:** SIH, NASA Space App Challenge, Hackout'26, ET Gen AI, Arbitrage Arena, Ingenium